SignalRisk

User Guide

Version 2.1.0

March 2026

For fraud analysts and operations teams

Confidential

SignalRisk Dashboard — User Guide

> For fraud analysts and operations teams | Verified baseline: 11 March 2026

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1. Getting Started

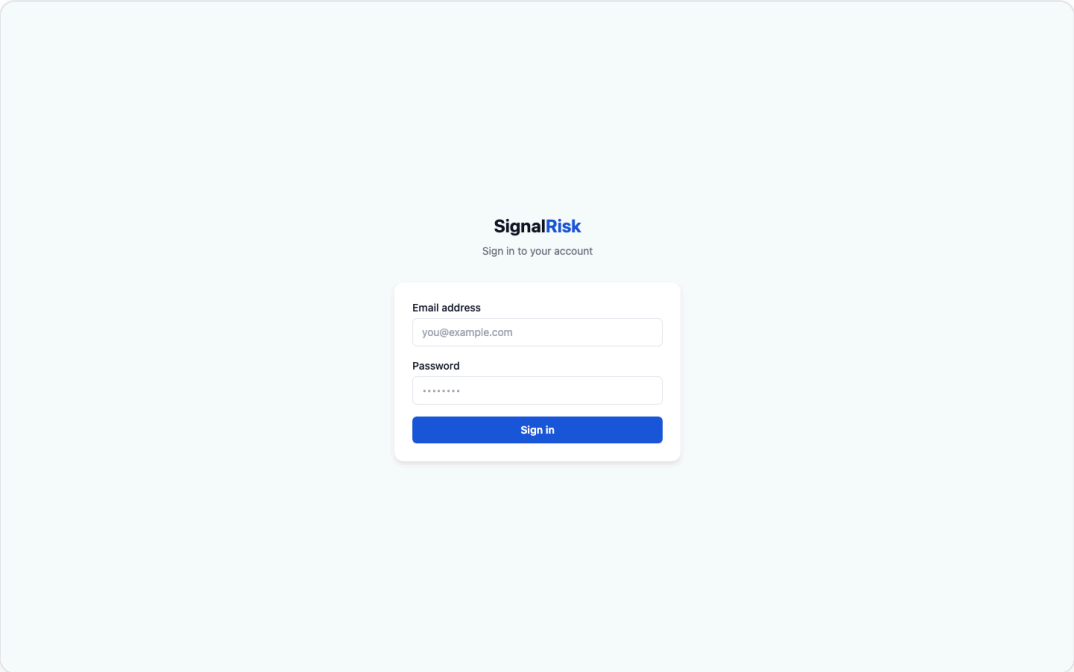

Figure 1 — Login Page

What is SignalRisk?

SignalRisk is a real-time fraud decision engine that evaluates incoming transactions against multiple intelligence signals — device fingerprinting, behavioural analysis, network reputation, velocity tracking, telco intelligence, and graph-based entity analysis — to produce instant ALLOW, REVIEW, or BLOCK decisions.

The Dashboard is the analyst-facing web application for monitoring fraud activity, managing cases, tuning detection rules, and running adversarial tests against the detection engine.

Logging In

Navigate to your dashboard URL (`http://localhost:5173` in development) and enter your credentials.

Current verified behavior:

- dashboard login uses seed users in development
- seed-user login is disabled when `NODE_ENV=production`
- the dashboard stores auth state client-side and redirects to the Overview page after successful login

Development seed users:

- **Admin account**: `admin@signalrisk.io`
- **Analyst account**: `analyst@signalrisk.io`

Default development passwords are environment-driven and may fall back to seed values in local environments. Do not reuse seed credentials outside development.

After login you are taken directly to the **Overview** page.

Navigation

The left sidebar contains links to all pages, organized into two sections:

Main navigation: - Overview, Cases, Rules, Fraud Ops, Analytics, Graph Intel, Live Feed, Settings, Admin

Fraud Tester section: - Overview, Battle Arena, Scenarios, Reports, Agents, Targets

Your role badge and email are shown in the top-right corner. Use **Sign out** to end your current dashboard session.

The footer shows the platform version (currently v0.1.0).

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2. Overview Page

Path: `/` (home)

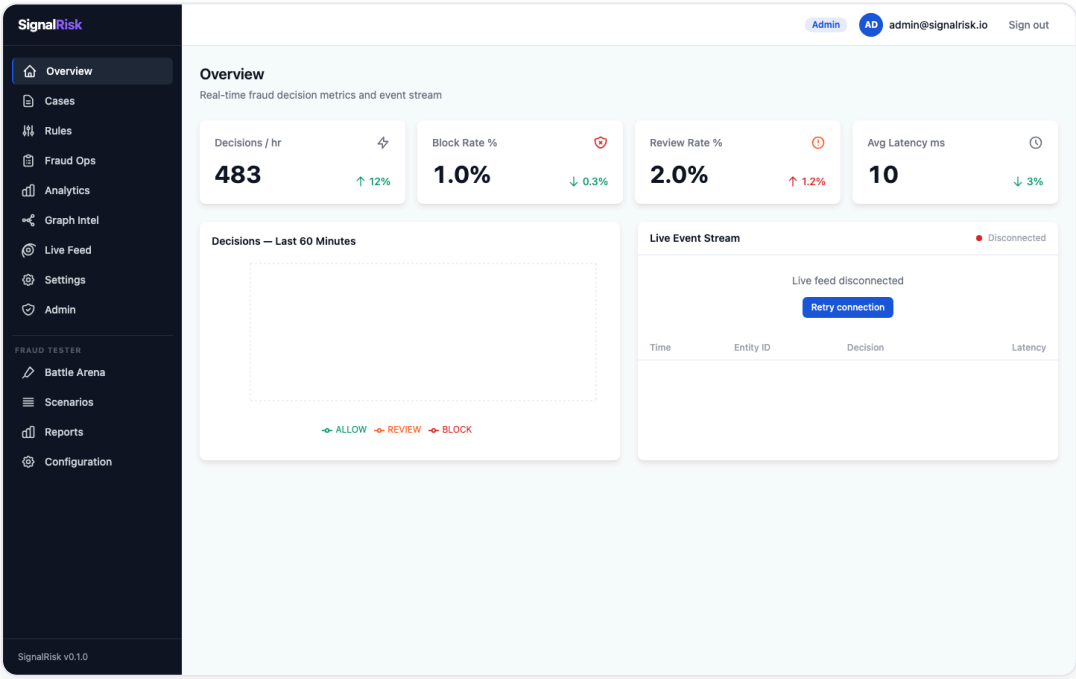

Figure 2 — Overview Dashboard

The Overview page provides a real-time snapshot of platform health and fraud activity.

KPI Cards

Four metric cards at the top update automatically every 30 seconds:

Card	Description
Decisions / hr	Number of fraud decisions made in the last hour
Block Rate %	Percentage of decisions that resulted in BLOCK
Review Rate %	Percentage of decisions that resulted in REVIEW
Avg Latency ms	Average decision processing time in milliseconds

Each card shows a trend arrow (up/down in green or red) and percentage change compared to the previous period. A yellow **Stale** badge appears if the data could not be refreshed.

Decision Trend Chart

The “Decisions — Last 60 Minutes” line chart shows decision counts broken down by outcome (ALLOW / REVIEW / BLOCK) at 2-minute intervals. The three lines are colour-coded: green (ALLOW), amber (REVIEW), red (BLOCK). Hover over any point to see exact counts.

Live Event Stream

The right panel shows the most recent decisions arriving in real time via WebSocket. Each row displays: - Timestamp - Entity ID (device or user fingerprint) - Decision outcome badge (colour-coded) - Processing latency in milliseconds

A green **Connected** indicator confirms the WebSocket is active. If it shows red **Disconnected**, click **Retry connection** to reconnect.

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3. Cases Page

Path: `/cases`

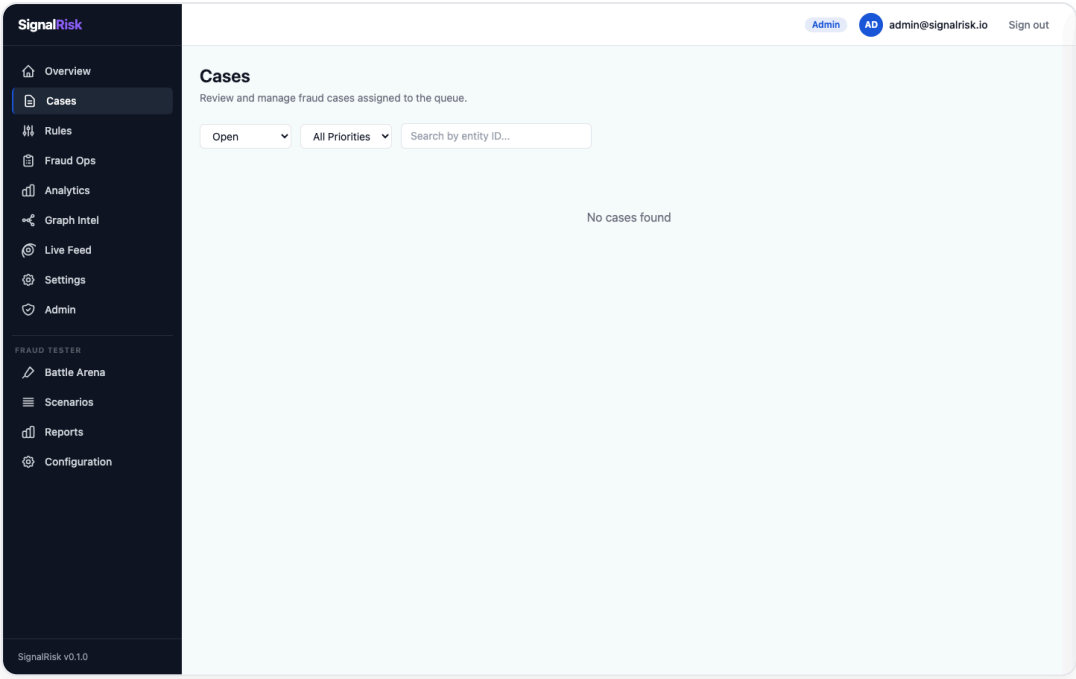

Figure 3 — Cases Queue

The Cases page is the primary analyst workflow queue. It shows all cases that require human review.

Understanding the Queue

By default the queue shows only **Open** cases — cases awaiting a decision. Cases automatically leave the queue when you Resolve or Escalate them.

Each row in the table shows: - **Case ID** — unique identifier - **Entity ID** — device fingerprint or user ID that triggered the event - **Action** — the automated decision (REVIEW or BLOCK) - **Priority** — HIGH / MEDIUM / LOW based on risk score - **Risk Score** — numeric score 0–100 - **SLA** — remaining time before SLA breach - **Status** — current case status - **Assignee** — analyst assigned to the case - **Actions** — View button

Filtering

Use the filter bar at the top to narrow the queue:

- **Status dropdown** — Open / In Review / Escalated / Resolved
- **Priority dropdown** — All Priorities / HIGH / MEDIUM / LOW
- **Search** — filter by Entity ID (300ms debounce)

Case Detail Panel

Click **View** on any row to open the sliding detail panel:

- **Decision Summary**: action, risk score, entity ID, priority
- **Risk Factors**: signal contributions with visual bars and descriptions
- **SLA**: remaining time with colour indicator
- **Evidence Timeline**: sequence of events (received, scored, case created)
- **Resolution**: select FRAUD / LEGITIMATE / INCONCLUSIVE, add notes, submit
- **Escalate Case**: escalate to senior analyst

Pagination

Previous/Next buttons appear when there are more than 20 cases.

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4. Rules Page

Path: `/rules`

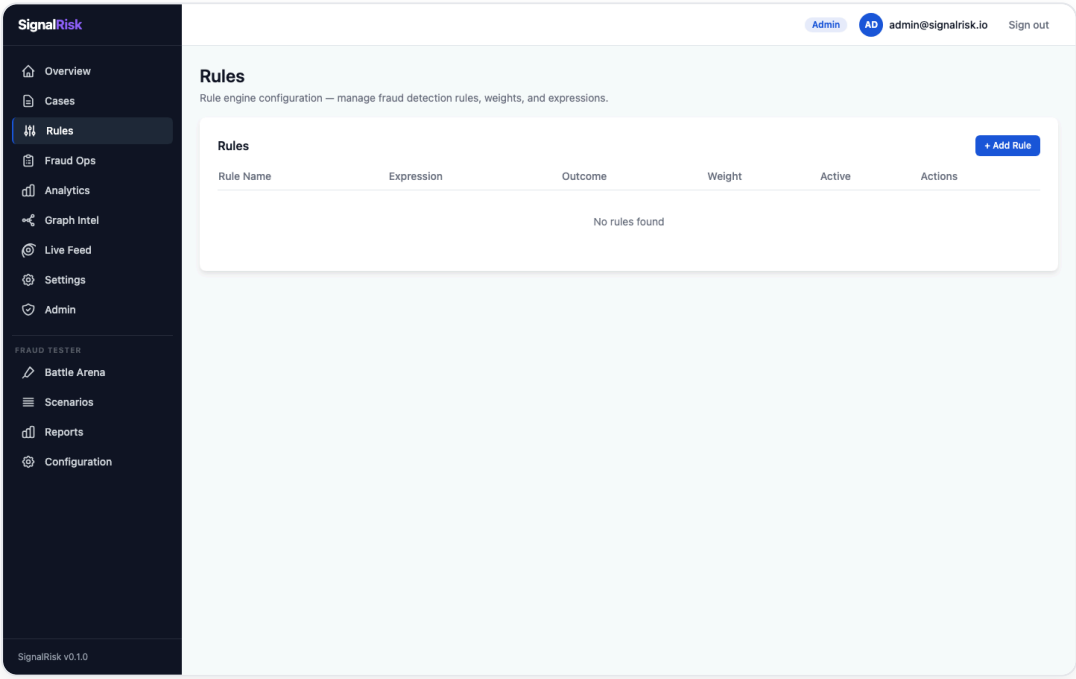

Figure 4 — Rules Management

The Rules page lets you manage the DSL rules that the decision engine evaluates against every incoming event.

Rules Table

Each rule shows:

Column	Description
Rule Name	Human-readable identifier
Expression	DSL condition (e.g. `velocity.burstDetected == true`, `stateful.customer.previousBlockCount30d > 2`)
Outcome	ALLOW, REVIEW, or BLOCK (colour-coded)
Weight	Contribution weight 0.10–1.00
Active	Yes/No toggle
Actions	Edit / Delete buttons

Adding a Rule

Click **+ Add Rule** to open the modal:

1. **Rule Name** — descriptive name
2. **DSL Expression** — condition using contexts: `device`, `velocity`, `behavioral`, `network`, `telco`, `txn`, `stateful` (including `stateful.customer.*`, `stateful.device.*`, `stateful.ip.*`, `stateful.graph.*`, `stateful.sequence.*`)
3. **Outcome** — ALLOW, REVIEW, or BLOCK
4. **Weight** — initial contribution (0.10–1.00)
5. **Active immediately** — check to enable on creation

Toggling and Weight Adjustment

- Click the Active column to instantly enable/disable a rule (no deploy needed)
- Drag the Weight slider to change rule influence on the final score

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5. Fraud Ops Page

Path: `/fraud-ops`

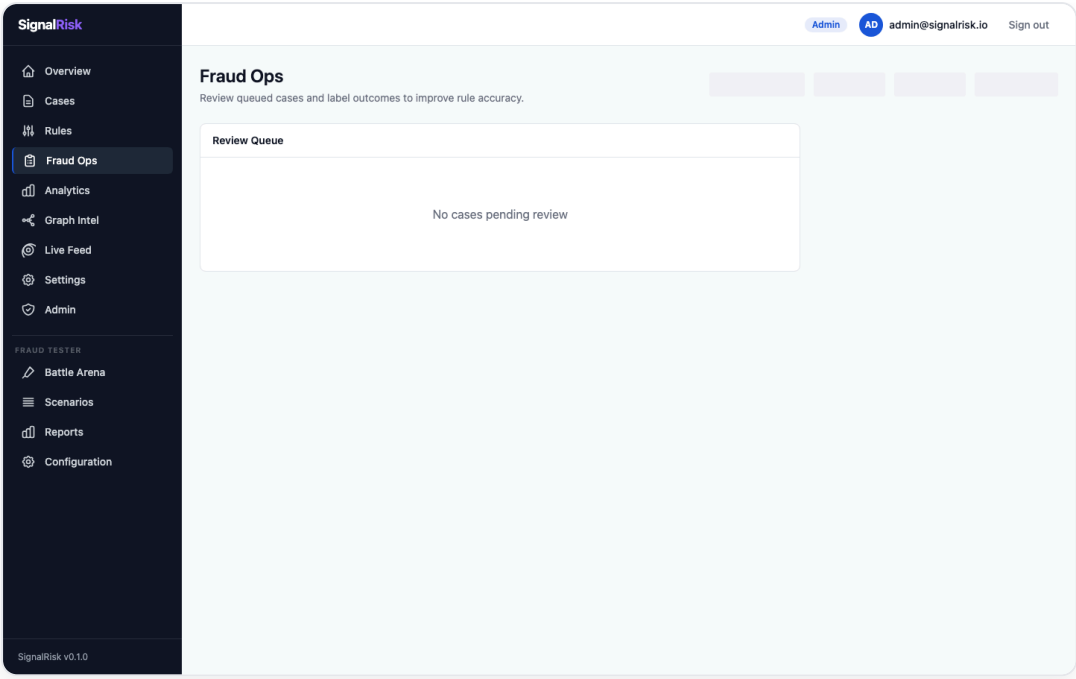

Figure 5 — Fraud Ops Labeling

The Fraud Ops page is for labelling cases to improve the accuracy of the rule engine. Unlike the Cases page (which manages the queue), Fraud Ops focuses on REVIEW-action cases specifically for training purposes.

Labeling Stats Bar

Top-right stats show today's labeling activity: - **Today: N labeled** — total decisions made today - **Fraud: N** — confirmed fraud labels - **FP: N** — false positive (legitimate) labels - **Accuracy: N%** — fraud confirmed / (fraud + false positive) - **Pending: N** — cases awaiting labeling

Review Queue

The "Review Queue" lists REVIEW-action cases sorted by risk score (highest first). When no cases are pending, it shows "No cases pending review".

Outcome Modal

Click **Label** to choose: - **Fraud Confirmed** — genuinely fraudulent (rule weight +0.05) - **False Positive** — legitimate event (rule weight −0.03) - **Inconclusive** — insufficient evidence

6. Analytics Page

Path: `/analytics`

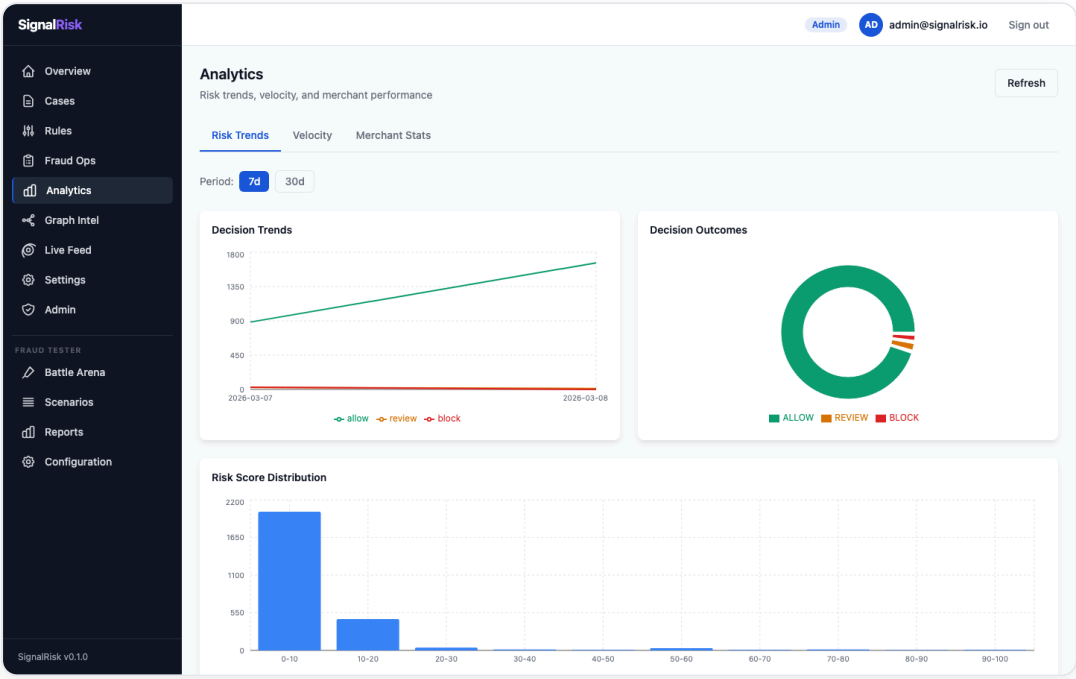

Figure 6 — Analytics: Risk Trends

The Analytics page provides historical performance metrics with three tabbed views and a period selector.

Risk Trends Tab

- **Period selector**: 7d / 30d toggle buttons
- **Decision Trends chart**: Line chart showing daily ALLOW/REVIEW/BLOCK counts over time (green/amber/red lines)
- **Decision Outcomes donut**: Pie/donut chart showing the proportion of ALLOW, REVIEW, and BLOCK decisions
- **Risk Score Distribution**: Bar chart showing how many decisions fall into each 10-point risk bucket (0-10, 10-20, ..., 90-100)

Velocity Tab

- **Events per Hour**: Area chart showing event volume over time, useful for identifying traffic spikes that may indicate an ongoing attack

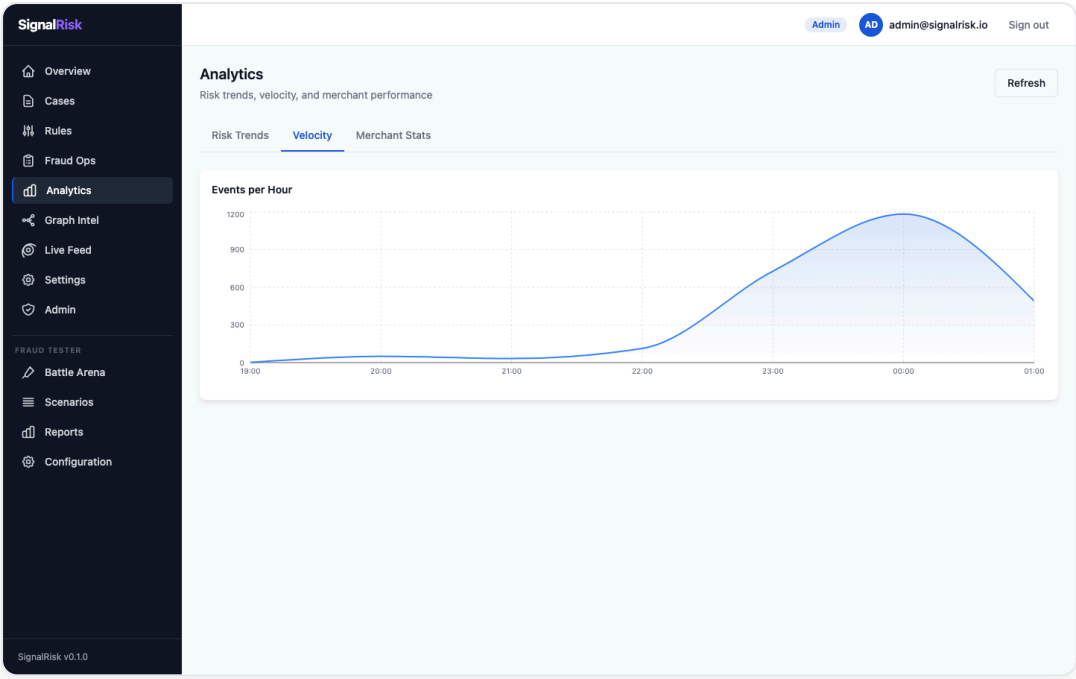

Figure 6b — Analytics: Velocity Tab

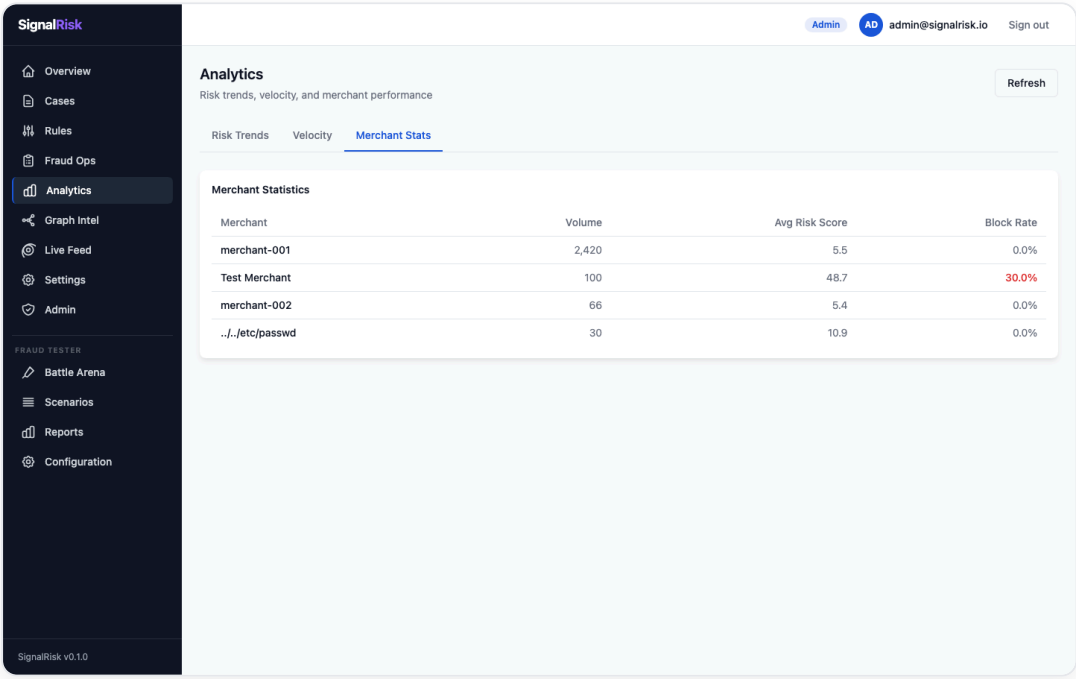

Figure 6c — Analytics: Merchant Stats Tab

Merchant Stats Tab

- **Merchant Statistics table**: Per-merchant breakdown showing:
  - Merchant name/ID
  - Volume (total events)
  - Avg Risk Score

- Block Rate % (highlighted in red when > 10%)

Use the **Refresh** button (top-right) to force-reload analytics data.

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7. Graph Intelligence Page

Path: `/graph-intel`

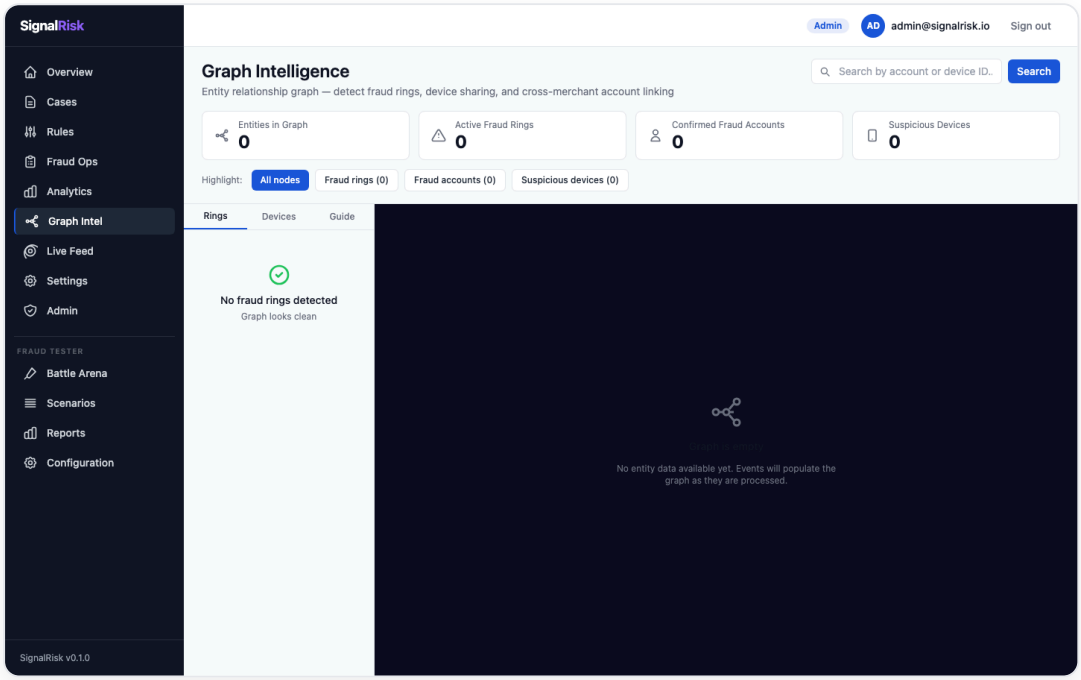

Figure 7 — Graph Intelligence

The Graph Intelligence page visualizes entity relationships to detect fraud rings, device sharing, and cross-merchant account linking.

KPI Cards

Four cards at the top:

Card	Description
Entities in Graph	Total nodes in the entity relationship graph
Active Fraud Rings	Detected fraud ring clusters
Confirmed Fraud Accounts	Accounts confirmed as fraudulent
Suspicious Devices	Devices flagged as suspicious

Highlight Filters

Toggle buttons to focus the graph visualization: - **All nodes** — show everything - **Fraud rings (N)** — highlight only fraud ring members - **Fraud accounts (N)** — highlight confirmed fraud accounts - **Suspicious devices (N)** — highlight suspicious devices

Detail Tabs

- **Rings** — lists detected fraud rings with “No fraud rings detected / Graph looks clean” when none exist
- **Devices** — lists suspicious devices
- **Guide** — usage guide for interpreting the graph

Search

Use the search bar (top-right) to search by account or device ID. Click **Search** to focus the graph on a specific entity.

Graph Visualization

The main area displays an interactive network graph. Nodes represent entities (devices, accounts), edges represent relationships (shared IP, shared device, etc.). The graph populates as events are processed through the system.

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8. Live Feed Page

Path: `/live-feed`

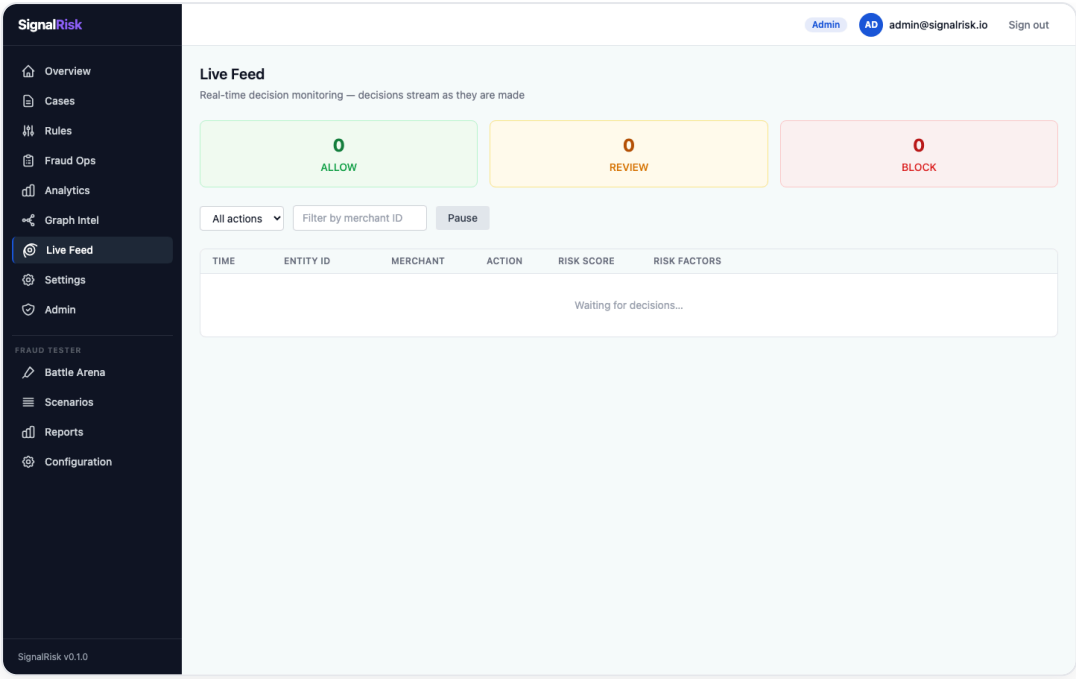

Figure 8 — Live Feed

The Live Feed page shows real-time decisions as they happen via WebSocket.

Counter Cards

Three colour-coded cards at the top show running totals: - **ALLOW** (green) — allowed decisions - **REVIEW** (amber) — review decisions - **BLOCK** (red) — blocked decisions

Controls

- **Action filter dropdown** — filter by All actions / ALLOW / REVIEW / BLOCK
- **Merchant ID filter** — filter by specific merchant
- **Pause button** — pause/resume the live stream

Decision Table

Column	Description
TIME	When the decision was made
ENTITY ID	Device/user fingerprint
MERCHANT	Merchant ID
ACTION	ALLOW / REVIEW / BLOCK (colour-coded)
RISK SCORE	Numeric risk score
RISK FACTORS	Contributing signals

Use this page on a monitoring screen during high-risk periods or incident response.

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9. Settings Page

Path: `/settings`

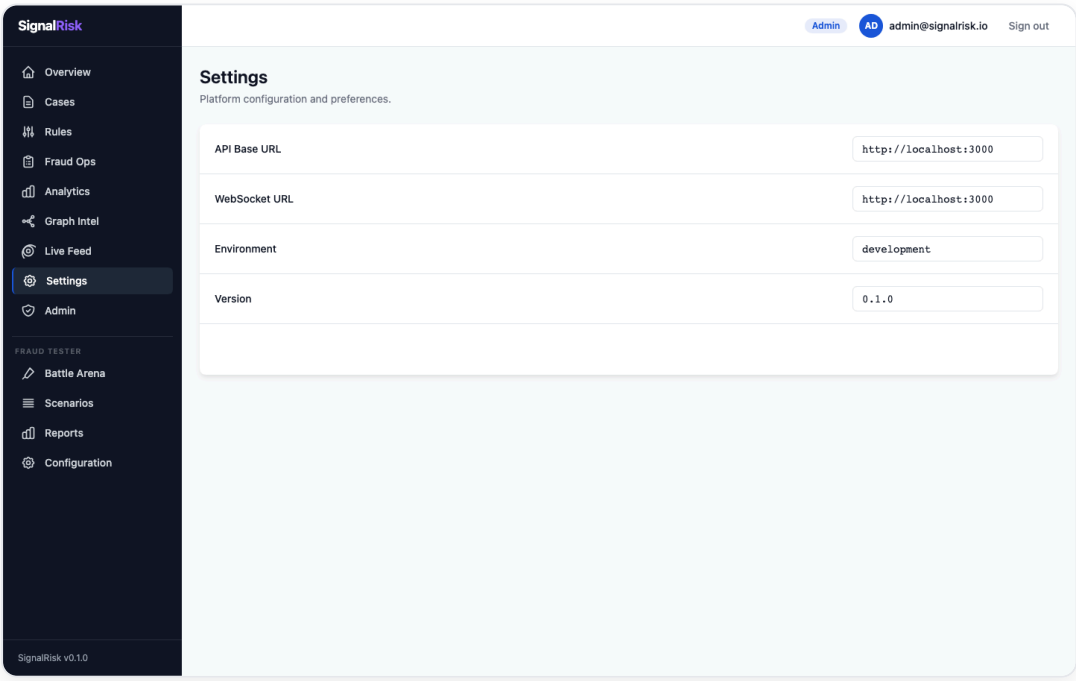

Figure 9 — Settings

The Settings page is a lightweight local preferences form. It does not manage server-side platform configuration.

Current verified fields:

Setting	Description	Current Default
API Base URL	Stored local API base URL preference	`http://localhost:3000`
WebSocket URL	Live feed WebSocket endpoint	`http://localhost:3000`
Environment	Current environment	`development`
Version	Platform version	`0.1.0`

Settings are persisted to `localStorage` and survive page reloads.

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10. Admin Page

Path: `/admin` (Admin role required)

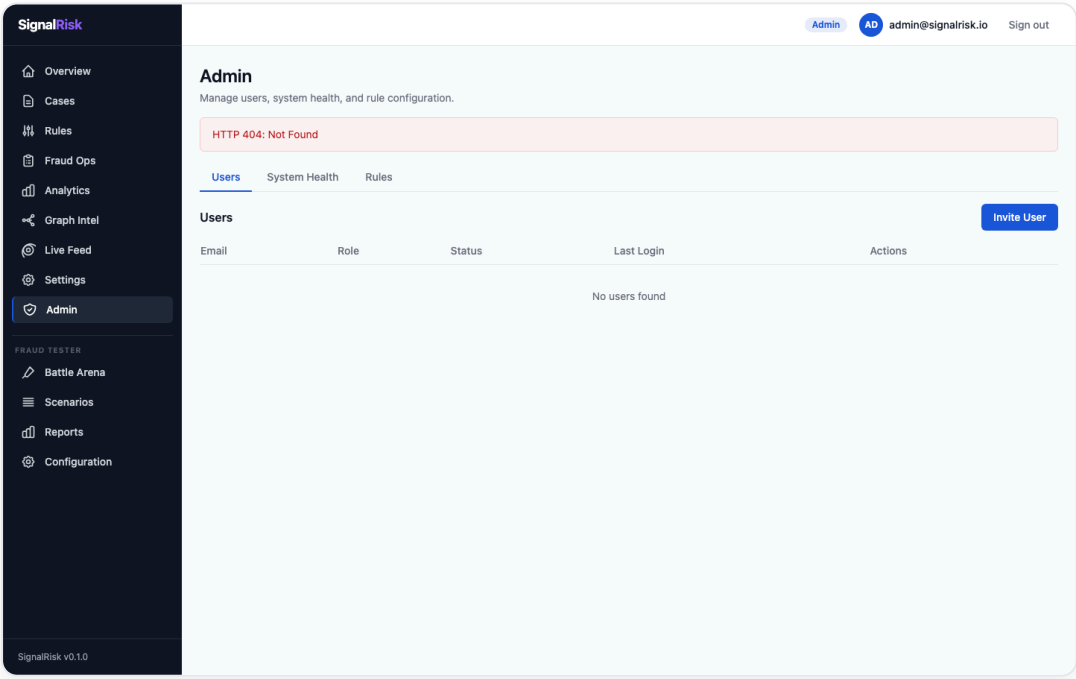

Figure 10 — Admin Panel

The Admin page is organized into three tabs:

Users Tab

- Lists users available through the admin API/store
- Includes user management actions surfaced by the current admin UI
- Use this area for operational user administration, not merchant auth onboarding

System Health Tab

- Shows live health status of all microservices
- **All Systems Operational** banner (green) when everything is healthy
- Service list with status indicators (healthy/degraded/down)
- **Refresh** button and "Last checked" timestamp

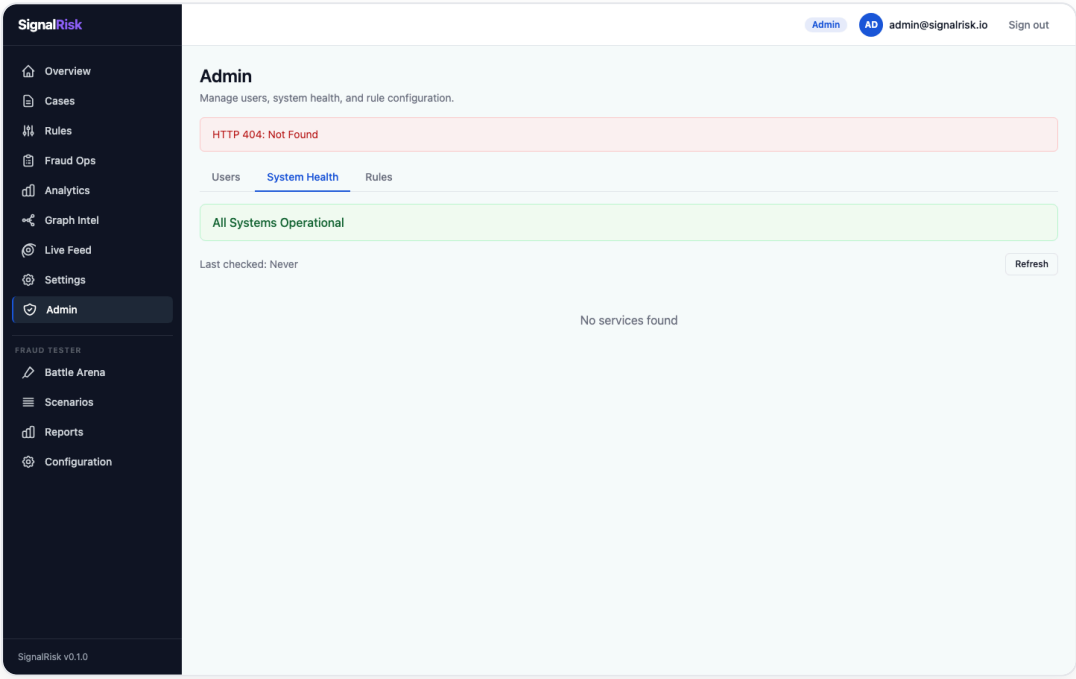

Figure 10b — Admin: System Health Tab

Rules Tab

Same as the Rules page — provides full rule CRUD functionality.

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11. FraudTester — Battle Arena

Path: `/fraud-tester/battle-arena`

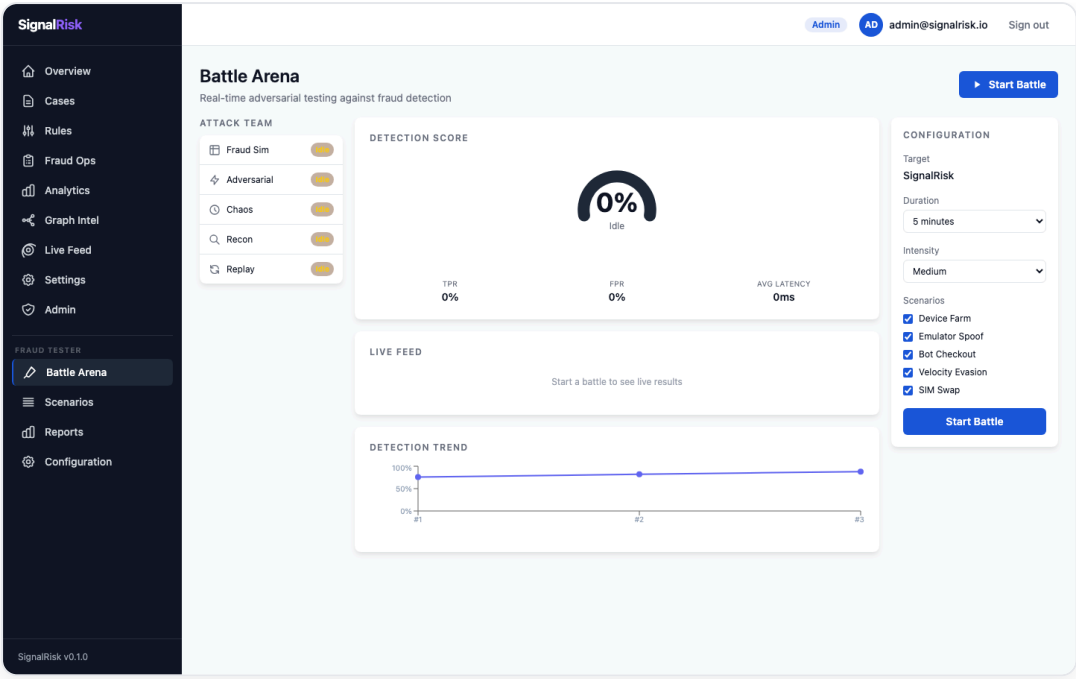

Figure 11 — FraudTester Battle Arena

What is FraudTester?

FraudTester is a **QA and Red Team tool** for testing SignalRisk's fraud detection capabilities. It generates synthetic fraud events, submits them through the real pipeline, and measures detection accuracy.

Who uses it: Security/QA engineers, fraud analysts, developers **What it measures:** True Positive Rate (TPR), False Positive Rate (FPR), latency

Layout

The page is divided into three sections:

Left — Attack Team: Shows the 5 AI agents with status badges: - Fraud Sim — standard fraud pattern simulation - Adversarial — adaptive evasion techniques - Chaos — random noise and edge cases - Recon — reconnaissance patterns - Replay — replay attacks

Center — Detection Score: - Detection gauge (0-100%) with Idle/Running state - TPR / FPR / AVG LATENCY metrics - LIVE FEED showing real-time attack results - DETECTION TREND chart tracking detection rate across battles

Right — Configuration: - **Target**: Select target system (SignalRisk) - **Duration**: 5 minutes / 10 minutes / 30 minutes - **Intensity**: Low / Medium / High - **Scenarios**: Checkboxes for Device Farm, Emulator Spoof, Bot Checkout, Velocity Evasion, SIM Swap

Starting a Battle

1. Configure scenarios, duration, and intensity in the right panel
2. Click **Start Battle** (blue button in header or config panel)
3. Watch real-time results in the center panel
4. Results are colour-coded: red (BLOCK = detected), amber (REVIEW), green (ALLOW = missed)

Test Isolation

All Battle Arena traffic is automatically marked with `X-SignalRisk-Test: true`. Test data is excluded from production analytics and no webhooks are triggered.

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12. FraudTester — Scenario Library

Path: `/fraud-tester/scenarios`

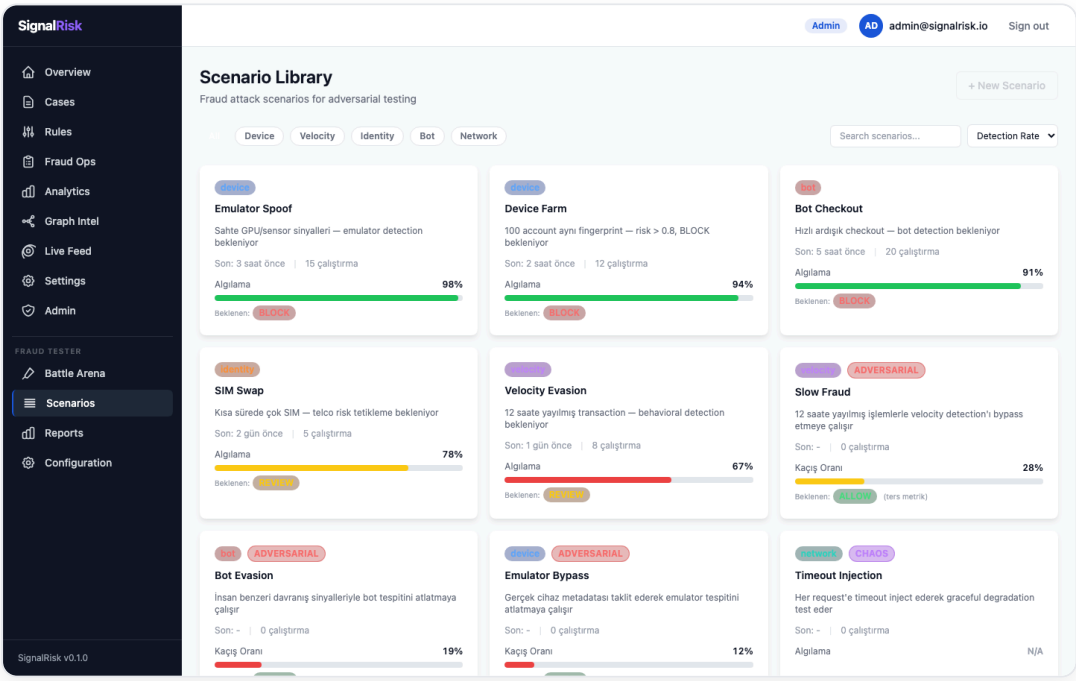

Figure 12 — Scenario Library

The Scenario Library displays all available fraud test scenarios.

Category Filters

Filter buttons at the top: **Device**, **Velocity**, **Identity**, **Bot**, **Network**

Sort and Search

- **Search** field for filtering by name
- **Sort by** dropdown (Detection Rate)

Scenario Cards

Each card shows:

Field	Description
Category badges	Colour-coded tags (device, velocity, bot, identity, network, adversarial, chaos)
Scenario name	e.g. Emulator Spoof, Device Farm, Bot Checkout
Description	What the attack pattern does
Last run	When the scenario was last executed
Run count	Total number of executions
Detection rate	Historical detection percentage (bar + %)
Expected outcome	BLOCK / REVIEW / ALLOW badge

Available Scenarios (9+)

Scenario	Category	Detection Rate	Expected
Emulator Spoof	device	98%	BLOCK
Device Farm	device	94%	BLOCK
Bot Checkout	bot	91%	BLOCK
SIM Swap	identity	78%	REVIEW
Velocity Evasion	velocity	67%	REVIEW
Slow Fraud	velocity + adversarial	28% evasion	ALLOW
Bot Evasion	bot + adversarial	—	Tests limits
Emulator Bypass	device + adversarial	—	Tests limits
Timeout Injection	network + chaos	—	Graceful degradation

+ New Scenario

Click **+ New Scenario** (top-right) to create a custom test scenario.

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13. FraudTester — Detection Reports

Path: `/fraud-tester/reports`

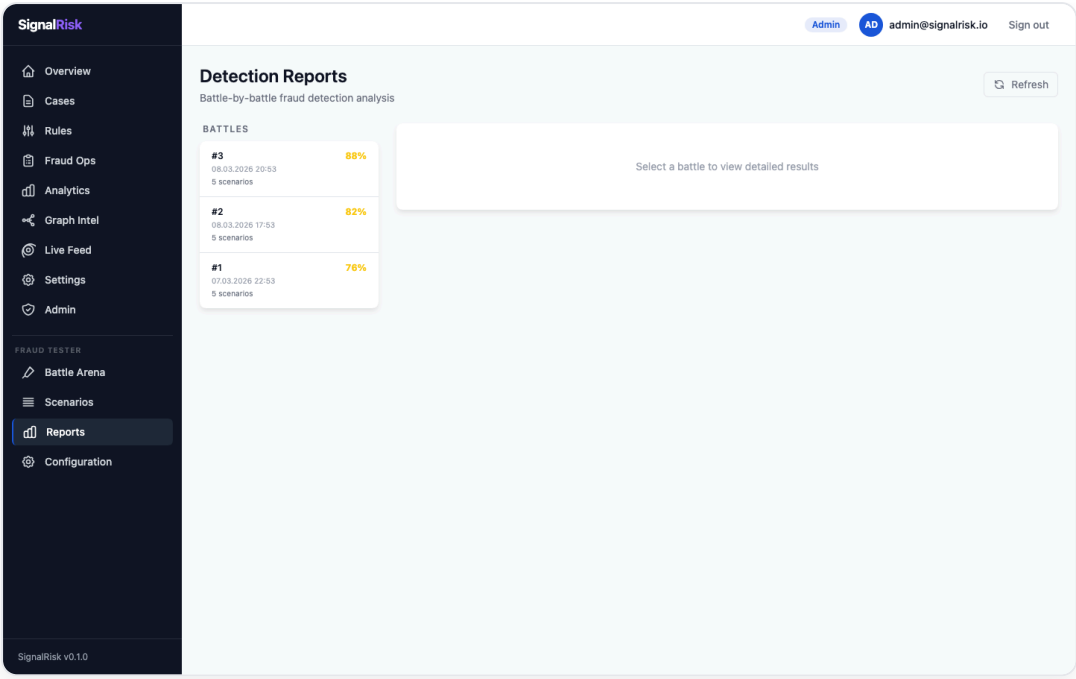

Figure 13 — Detection Reports

The Detection Reports page shows historical battle results and detection performance trends.

Battle List

Left panel lists all completed battles: - Battle number (#1, #2, #3...) - Date and time - Scenario count - Overall detection rate % (colour-coded)

Battle Detail (right panel)

Click a battle to view detailed results:

KPI Cards: - TPR (True Positive Rate) - FPR (False Positive Rate) - AVG LATENCY (per decision) - SCENARIOS RUN (passed count)

Scenario Breakdown Table:

Column	Description
SCENARIO	Scenario name
DETECTION	Detection rate with progress bar
ESCAPED (FN)	False negatives (e.g. 1/10)
AVG LATENCY	Average processing time
STATUS	PASS / FAIL badge

Trend — Last 3 Battles: - Detection Rate Trend (TPR) line chart across battles - Avg Latency per Battle bar chart

Interpreting Results

- **TPR > 85%** is the target for production readiness
- **FPR < 5%** ensures legitimate users are not blocked
- Rising FPR → overly aggressive rules, need weight adjustment
- Falling TPR on adversarial scenarios → detection gaps to address

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14. FraudTester — Agent Configuration

Path: `/fraud-tester/agents` (sidebar: Configuration)

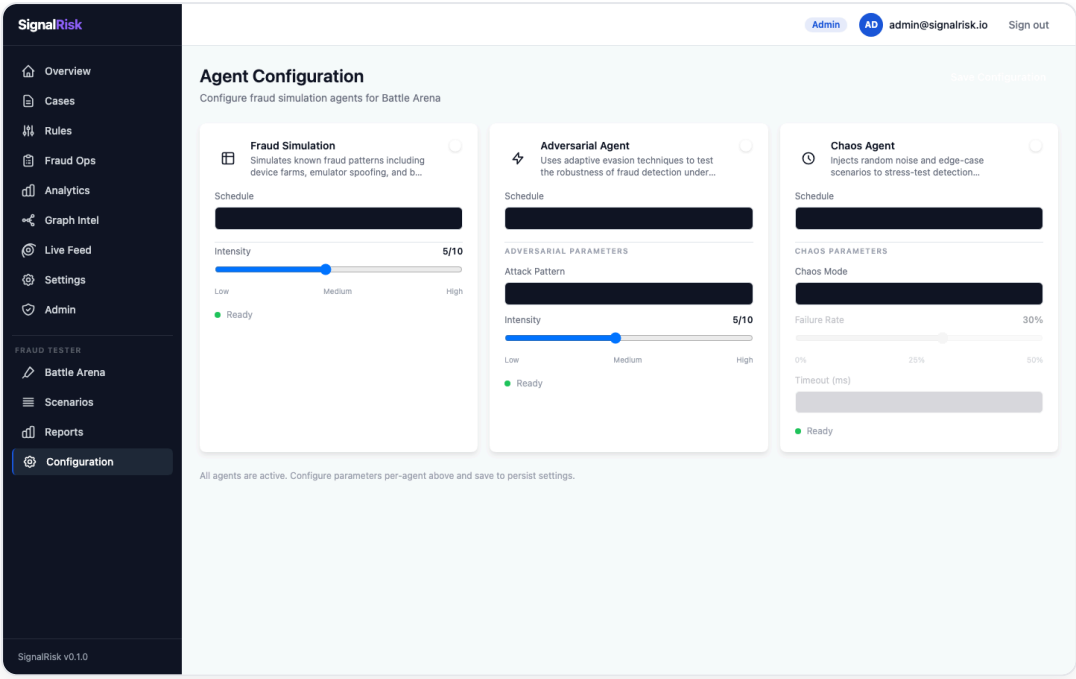

Figure 14 — Agent Configuration

The Agent Configuration page lets you configure the three AI agents used in Battle Arena.

Fraud Simulation Agent

- **Description**: Simulates known fraud patterns including device farms, emulator spoofing, and bot checkout
- **Schedule**: Configurable schedule field
- **Intensity**: Slider 1-10 (Low / Medium / High)
- **Status**: Ready (green dot) / Running / Error

Adversarial Agent

- **Description**: Uses adaptive evasion techniques to test the robustness of fraud detection
- **Attack Pattern**: Configurable attack pattern field
- **Intensity**: Slider 1-10
- **Status**: Ready / Running / Error

Chaos Agent

- **Description**: Injects random noise and edge-case scenarios to stress-test detection resilience
- **Chaos Mode**: Configurable mode field

- **Failure Rate**: Slider 0-50% (default: 30%)
- **Timeout (ms)**: Configurable timeout field
- **Status**: Ready / Running / Error

All agents show a green “Ready” status indicator when properly configured. The footer message reads: “All agents are active. Configure parameters per-agent above and save to persist settings.”

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15. Roles & Permissions

Feature	Admin	Analyst	Merchant
Overview	Read	Read	Read (own)
Cases — view queue	Yes	Yes	Own only
Cases — resolve/escalate	Yes	Yes	No
Fraud Ops — label	Yes	Yes	No
Rules — view	Yes	Yes	No
Rules — create/edit/delete	Yes	No	No
Analytics	Yes	Yes	Own only
Graph Intelligence	Yes	Yes	No
Live Feed	Yes	Yes	No
Admin panel	Yes	No	No
Settings	Yes	Yes	Yes
FraudTester — Battle Arena	Yes	Yes	No
FraudTester — Scenarios	Yes	Yes	No
FraudTester — Reports	Yes	Yes	No
FraudTester — Configuration	Yes	No	No

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16. Keyboard Shortcuts & Tips

Navigation Tips

- Press **Escape** or click the backdrop to close the Case Detail sliding panel
- Select **Open** in the status dropdown to return to the active queue after browsing resolved cases
- Sort by SLA column to work time-sensitive cases first (red = SLA breached)

Search

The search box in Cases filters by Entity ID. It debounces 300ms and cancels previous requests — type quickly without waiting.

Stale Data Badge

If Overview KPI cards show a yellow **Stale** badge, the API call failed. Data shown is from the last successful fetch. Retry happens on tab focus or after 30 seconds.

Rule Weight Best Practices

- Start new rules at **0.5 weight** and let the feedback loop adjust
- Only set **1.0** for rules with confirmed zero false positives
- Set **0.1** (minimum) rather than deleting — preserves history
- Deactivate rules before deleting to observe impact first

FraudTester Tips

- Use **low intensity** for initial rule validation, **high intensity** for stress testing
- Run adversarial scenarios after rule changes to verify detection
- Compare battle reports over time to track improvement
- All test data is automatically isolated from production — no cleanup needed
- Battle results do **not** appear in Dashboard analytics (test isolation)
- FraudTester = “how good is our detection?” / Dashboard = “what is happening right now?”

Analytics Tips

- Use the **7d / 30d** toggle to switch between short-term and long-term views

- Check Merchant Stats tab for high block rates that may indicate compromised merchants
- Velocity tab shows hourly event volume — useful for spotting attack patterns

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For API integration details, see `docs/dev/api-reference.md`. For technical architecture, see `docs/TECHNICAL.md`. For incident response, see `docs/runbooks/on-call-playbook.md`.